

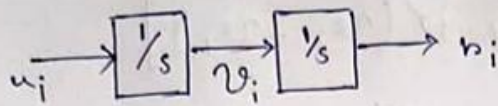
EE601

Leader-following consensus control of four double- integrator agents using the cooperative control

Aritra Biswas
Mainak Bhattacharyya
Arun E S
Manish Gupta

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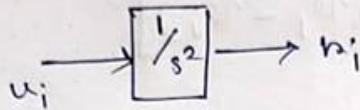


$$\dot{h}_i = v_i$$

$$\dot{v}_i = u_i$$

$$h_i \in \mathbb{R}^m$$

$$v_i \in \mathbb{R}^m$$



$$\ddot{h}_i = u_i$$

$$\dot{h}_i = v_i$$

$$\dot{v}_i = u_i$$

$h_i^{th} \rightarrow \text{position}$

$v_i^{th} \rightarrow \text{velocity}$

$$\begin{bmatrix} \dot{h}_i \\ \dot{v}_i \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} h_i \\ v_i \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_i$$

$$\dot{x}_i = A x_i + B u_i \quad x_i \in \mathbb{R}^2$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\rightarrow u_i = - \sum_{j=1}^n a_{ij} [(h_i - h_j) + \gamma (v_i - v_j)], \quad i \in I_n$$

where a_{ij} is the (i,j) th entry of the weighted adjacency matrix A characterizing the interaction graph for h_i and v_i , and γ is a positive gain.

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$$u_i = - \sum_{j=1}^n a_{ij} [(b_i - b_j) + \gamma (v_i - v_j)]$$

$$u_1 = - \sum_{j=1}^n a_{1j} [(b_1 - b_j) + \gamma (v_1 - v_j)]$$

$$= - [a_{12} [(b_1 - b_2) + \gamma (v_1 - v_2)]$$

$$+ a_{13} [(b_1 - b_3) + \gamma (v_1 - v_3)]$$

$$+ a_{14} [(b_1 - b_4) + \gamma (v_1 - v_4)]]$$

$$= - [(a_{12} + a_{13} + a_{14}) (b_1 + \gamma v_1) - a_{12} (b_2 + \gamma v_2) \\ - a_{13} (b_3 + \gamma v_3) - a_{14} (b_4 + \gamma v_4)]$$

$$= - (a_{12} + a_{13} + a_{14}) (b_1 + \gamma v_1) + a_{12} (b_2 + \gamma v_2)$$

$$+ a_{13} (b_3 + \gamma v_3) + a_{14} (b_4 + \gamma v_4)$$

$$u_2 = a_{21} (b_1 + \gamma v_1) - (a_{21} + a_{23} + a_{24}) (b_2 + \gamma v_2)$$

$$+ a_{23} (b_3 + \gamma v_3) + a_{24} (b_4 + \gamma v_4)$$

by continuing this process

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} -(a_{12} + a_{13} + a_{14}) & a_{12} & a_{13} & a_{14} \\ a_{21} & -(a_{21} + a_{23} + a_{24}) & a_{23} & a_{24} \\ a_{31} & a_{32} & -(a_{31} + a_{32} + a_{34}) & a_{34} \\ a_{41} & a_{42} & a_{43} & -(a_{41} + a_{42} + a_{43}) \end{bmatrix}$$

4×4

$$\begin{bmatrix} 1 & \gamma & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \gamma & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \gamma & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \gamma \end{bmatrix} \begin{bmatrix} h_1 \\ u_1 \\ h_2 \\ u_2 \\ h_3 \\ u_3 \\ h_4 \\ u_4 \end{bmatrix}$$

4×8 8×1

$\Downarrow$

$I_4 \otimes [1 \ \gamma]$

$\left(I_4 \otimes [1 \ \gamma] \right)$

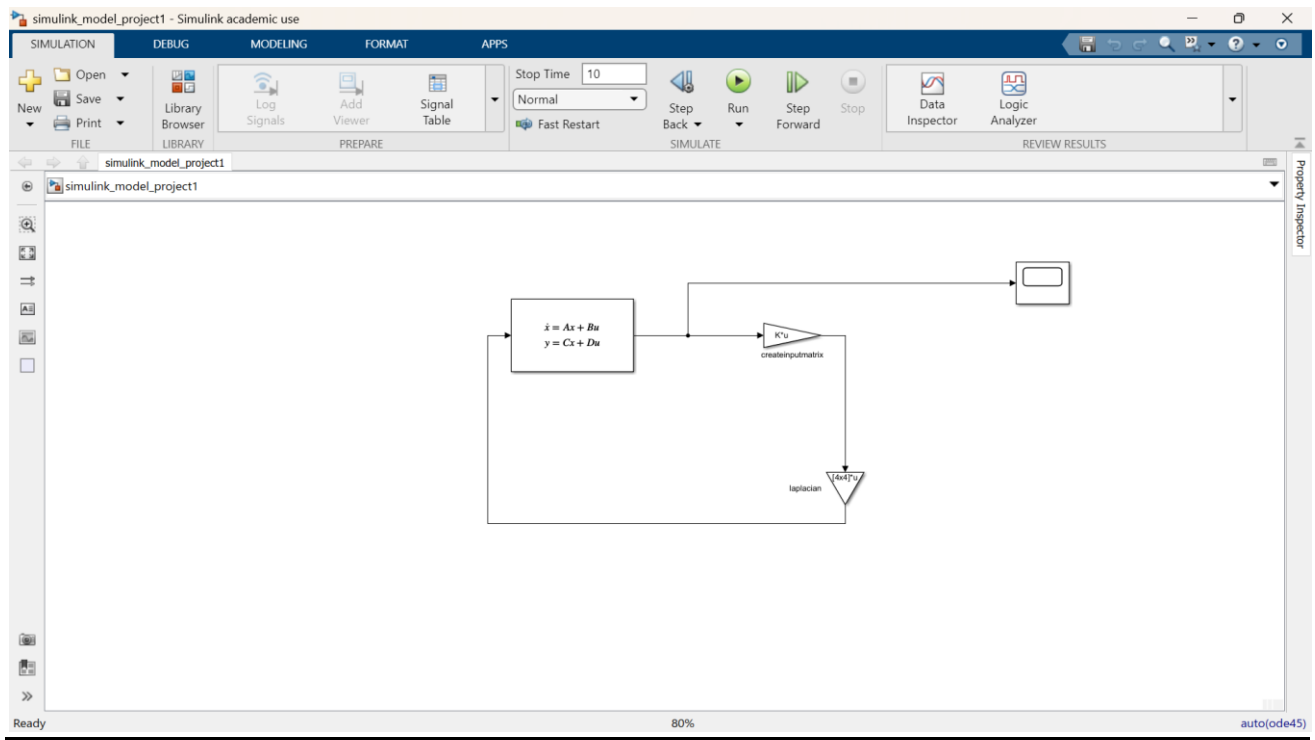
$\Downarrow$

4×4 1×2

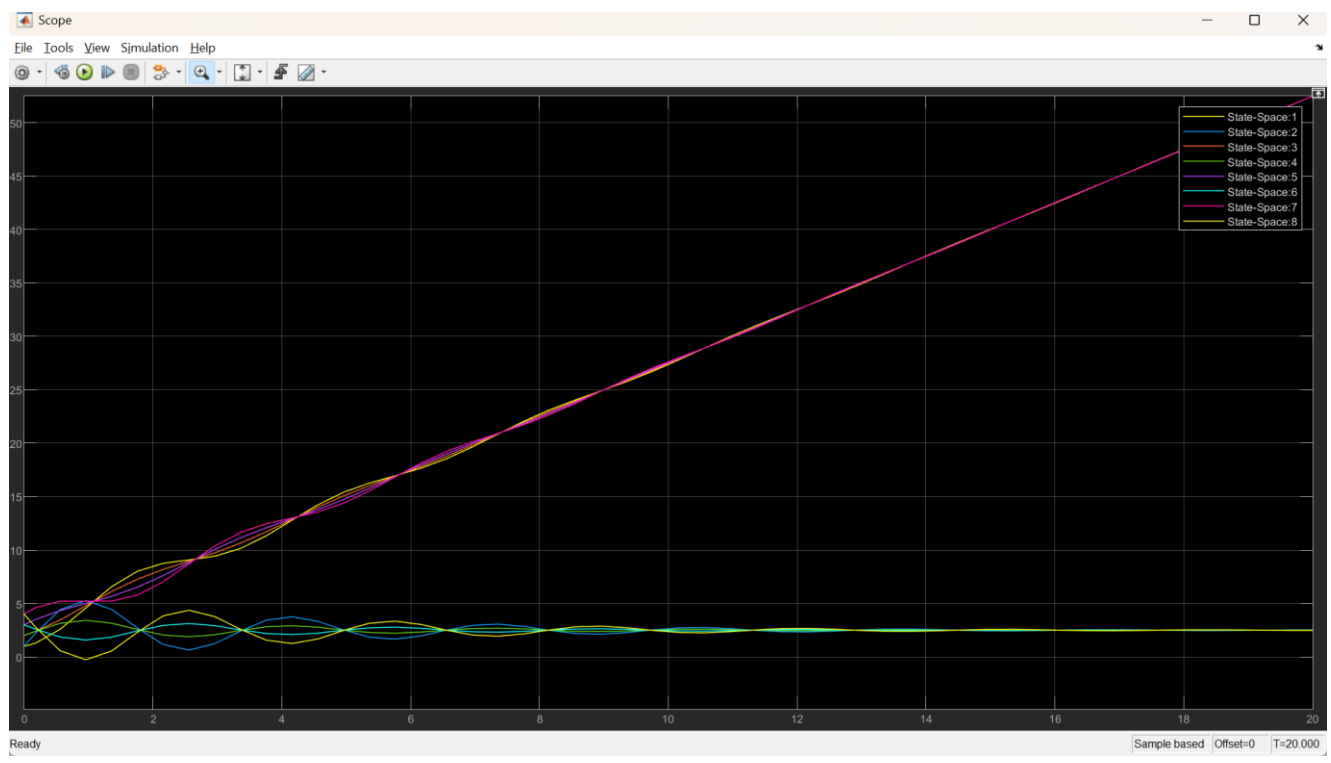
$$\bar{u} = -\mathcal{L} (I_4 \otimes [1 \ \gamma]) \bar{x}$$

MATLAB SIMULATION

Block Diagram:



Result:



in the presence of a group reference velocity $v^d \in \mathbb{R}^m$, a consensus algorithm for u_i is proposed in as:

$$u_i = \dot{v}^d - \alpha(v_i - v^d) - \sum_{j=1}^n a_{ij} [(b_i - b_j) + \gamma(v_i - v_j)]$$

$i \in \mathbb{I}_n$

where a_{ij} is the (i, j) th entry of the weighted adjacency matrix A characterizing the interaction graph for b_i and v_i and α and γ are positive gains,

consensus is reached for

$$u_i = - \sum_{j=1}^n a_{ij} [(b_i - b_j) + \gamma(v_i - v_j)] \quad i \in \mathbb{I}_n$$

if for all $b_i(0)$ and $v_i(0)$

$b_i(t) \rightarrow b_j(t)$ and $v_i(t) \rightarrow v_j(t)$ asymptotically

as $t \rightarrow \infty$ consensus is reached for

$$u_i = \dot{v}^d - \alpha(v_i - v^d) - \sum_{j=1}^n a_{ij} [(b_i - b_j) + \gamma(v_i - v_j)]$$

if for all $b_i(0)$ and $v_i(0)$ $b_i(t) \rightarrow b_j(t)$

and $v_i(t) \rightarrow v^d(t)$ asymptotically as $t \rightarrow \infty$

For group velocity its a leader following problem ($\dot{r}_d = 0$) (as for our case constant velocity $r_d = \text{constant}$).

$$u_i = -\alpha(r_{e_i} - r_d) - \sum_{j=1}^n \alpha_{ij} [(h_i - h_j) + \gamma(r_{e_i} - r_{e_j})]$$

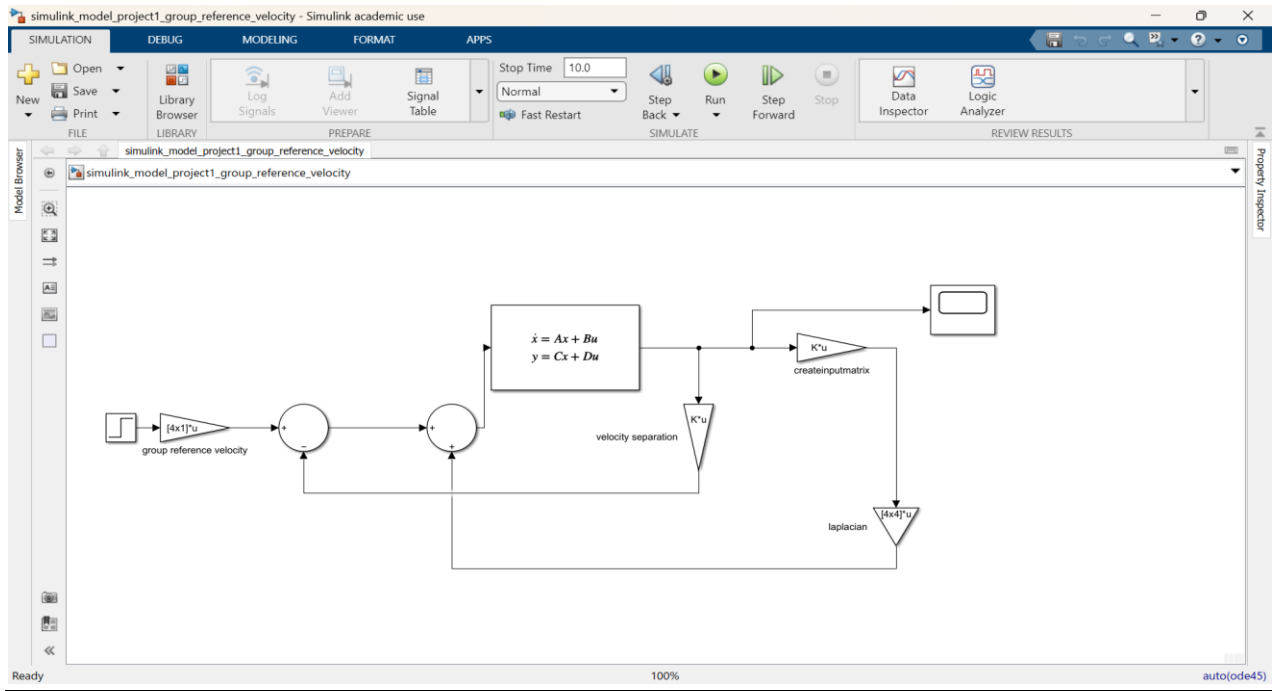
$$\bar{U} = -\alpha(\bar{r}_e - \bar{r}_d) - \mathcal{L} (I_4 \otimes [1 \ \gamma]) \bar{x}$$

$\bar{r}_d^d \in \mathbb{R}^4 \rightarrow$ group reference velocity

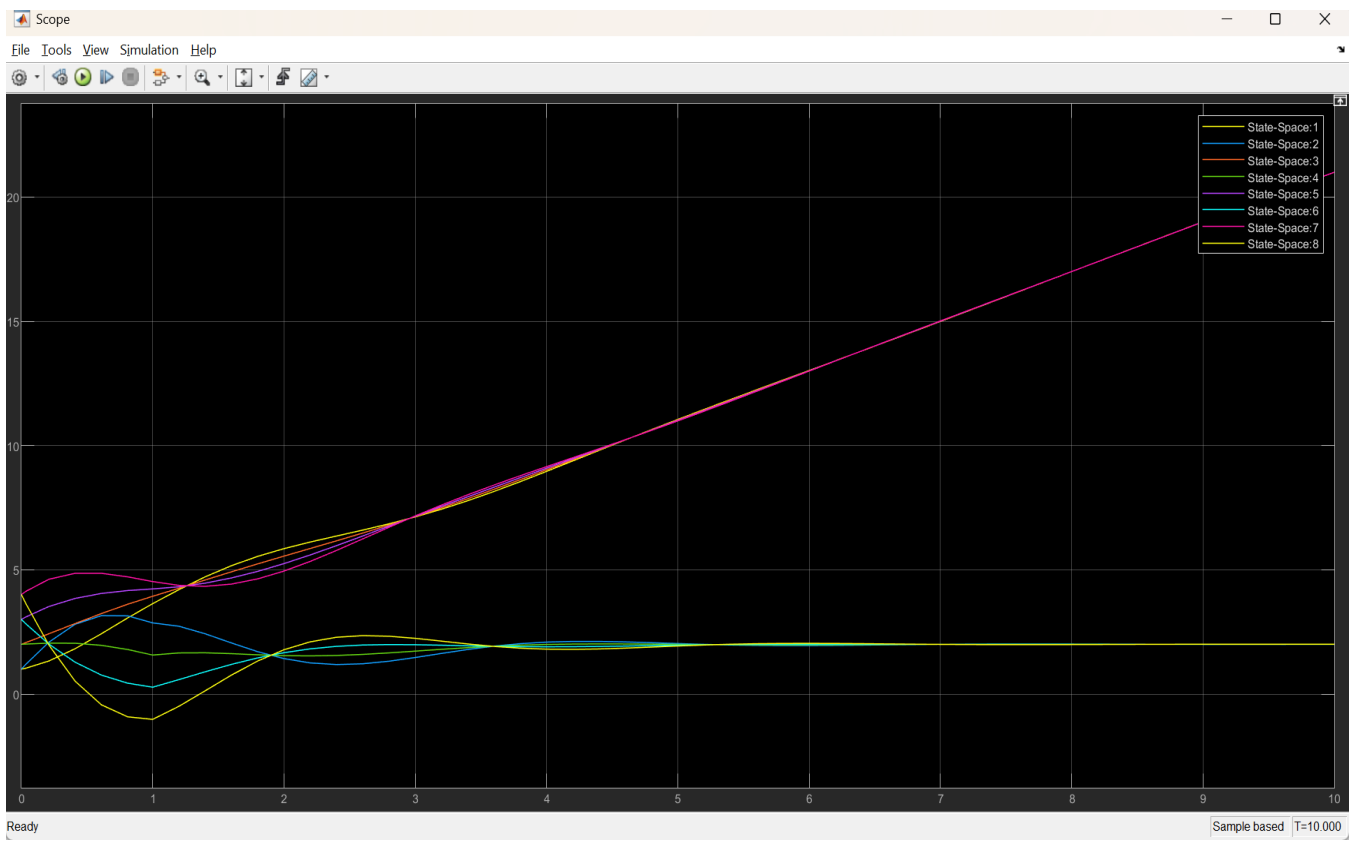
$$\bar{U} = \alpha(\bar{r}_d^d - \bar{r}_e) + (-\mathcal{L}) (I_4 \otimes [1 \ \gamma]) \bar{x}$$

MATLAB SIMULATION

Block Diagram:



Result:



Note that in

$$u_i = - \sum_{j=1}^n a_{ij} [(b_i - b_j) + r(r_{e_i} - r_{e_j})] \quad i \in \mathbb{I}_n$$

does not explicitly take into account actuator saturation. we propose a consensus algorithm for

$$\dot{b}_i = r_{e_i} \quad \text{and} \quad \dot{r}_{e_i} = u_i \quad i \in \mathbb{I}_n$$

with a bounded control input as

$$u_i = - \sum_{j=1}^n (a_{ij} \tanh[k_b(b_i - b_j)] + b_{ij} \tanh[k_r(r_{e_i} - r_{e_j})]) \quad i \in \mathbb{I}_n$$

where $K_b \in \mathbb{R}^{n \times n}$ and $K_r \in \mathbb{R}^{n \times n}$ are positive definite diagonal matrix a_{ij} and b_{ij} are respectively, the (i,j) th entry of the weighted adjacency matrix A and B characterizing, respectively, the interaction graphs for b_i and r_{e_i} and $\tanh(\cdot)$ is defined component-wise. that is,

$$\tanh([x_1, \dots, x_m]^T) = [\tanh(x_1), \dots, \tanh(x_m)]^T$$

where $\alpha_i \in \mathbb{R}$. Note that A and B can be chosen differently. Also note that with

$$u_i = - \sum_{j=1}^n (a_{ij} \tanh[k_b(b_i - b_j)] + b_{ij} \tanh[k_v(r_i - r_j)]) \quad i \in \mathbb{I}_n \quad (1)$$

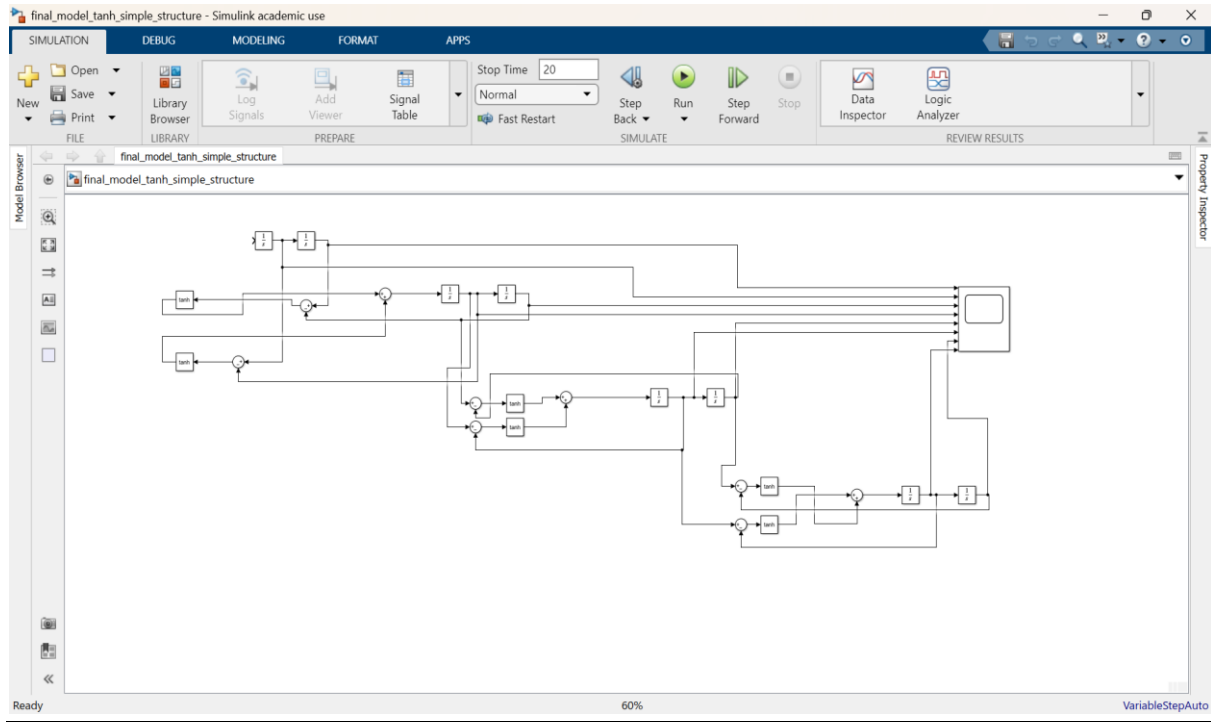
is bounded because $\tanh(\cdot)$ is bounded.

theorem

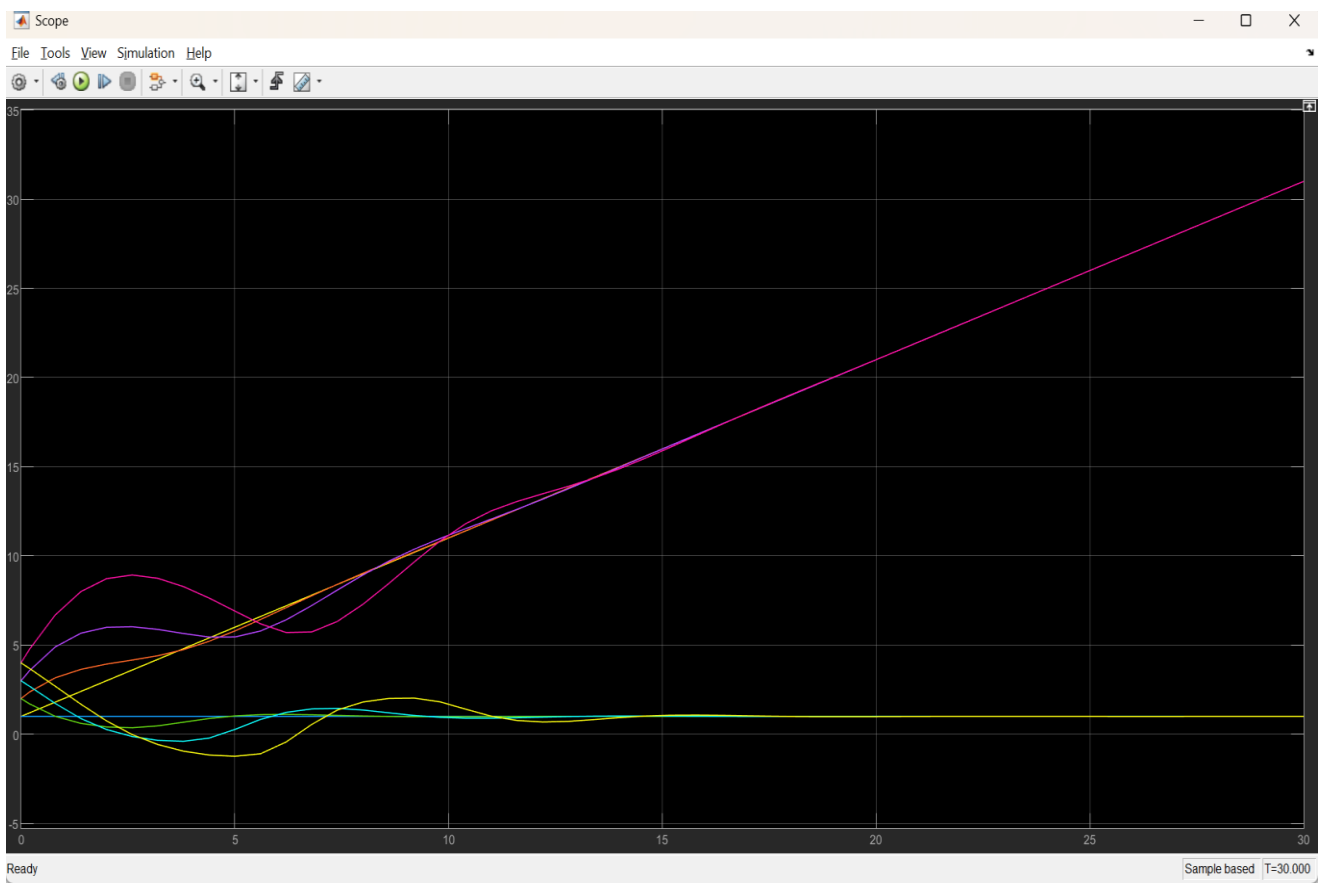
with (1), $b_i(t) \rightarrow b_j(t)$ and $r_i(t) \rightarrow r_j(t)$ asymptotically as $t \rightarrow \infty$ if the graph of A and B are undirected connected.

MATLAB SIMULATION

Block Diagram:



Result:



CONCLUSION

When the agents reach velocity consensus in the leaderless case, the common velocity is generally a nonzero constant, so each agent's absolute position grows linearly in time, while all relative position errors go to zero (i.e., $v_i - v_j = 0$, $v_i - v_j = 0$ for all $i \neq j$ and $i = j$).

For the first control law, the problem is leaderless: all four agents reach a common nonzero velocity, and their relative positions converge to zero.

For the second control law, there is a leader-following problem: all four agents track the prescribed group reference velocity, so each velocity converges to that reference value.

For the third control law, the problem is again leaderless: all four agents converge to a common weighted average of their initial velocities, and their positions converge to a (possibly different) weighted average position while still achieving consensus.